

5 Further calculus	5.1	Derive formulae for and calculate volumes of revolution.	Both $\pi \int y^2 dx$ and $\pi \int x^2 dy$ are required. Students should be able to find a volume of revolution given either Cartesian equations or parametric equations.
	5.2	Evaluate improper integrals where either the integrand is undefined at a value in the range of integration or the range of integration extends to infinity.	For example, $\int_0^{\infty} e^{-x} dx, \int_0^2 \frac{1}{\sqrt{x}} dx$
	5.3	Understand and evaluate the mean value of a function.	Students should be familiar with the mean value of a function $f(x)$ as, $\frac{1}{b-a} \int_a^b f(x) dx$
	5.4	Integrate using partial fractions.	Extend to quadratic factors $ax^2 + c$ in the denominator

Topic	What students need to learn:		
	Content		Guidance
5 Further calculus <i>continued</i>	5.5	Differentiate inverse trigonometric functions.	For example, students should be able to differentiate expressions such as, $\arcsin x + x\sqrt{1-x^2}$ and $\frac{1}{2}\arctan x^2$
	5.6	Integrate functions of the form $(a^2 - x^2)^{-\frac{1}{2}}$ and $(a^2 - x^2)^{-1}$ and be able to choose trigonometric substitutions to integrate associated functions.	